

Talk Like a Structured Graph: Augmenting Text Encodings with Structural Statistics

Inbal Moryles

github.com/iinbal
inbalmoryles@tau.ac.il

Gal Barak

github.com/GalBrk
galbarak2@mail.tau.ac.il

Nitzan Zacharia

github.com/NitzanZacharia
zacharia1@mail.tau.ac.il

Abstract

Does stating a graph’s structural statistics help a language model answer queries about a graph written as text? We prepend each node’s degree, local clustering or random-walk return probabilities to GraphQA’s incident encoding and compare the same 40-node graphs with and without them, for Qwen3-1.7B and Qwen3-4B with and without thinking, controlling for answer disclosure, added text and truncation. Stating the answer helps models that cannot compute it (+7.5 and +8.8 points on node degree) and costs one that can (−6.5); the models look the value up or check it, and the same values in a longer primer can cost accuracy. Other statistics have smaller effects whose sign depends on the statistic and the model; where structure-free text lowers accuracy, clustering and RWSE mostly do not, and they cut the non-edges a small model states as edges. On cycle detection only about a third of the correct answers given without thinking rest on a real cycle.

1 Introduction

Research question. When a black-box language model reads a textual graph, do explicit structural statistics improve execution of graph queries? By *execution* we mean computing the answer from the graph’s edges.

Graph models benefit from structural encodings, and how a graph is written as text changes an LLM’s accuracy on graph queries (§2); whether a frozen LLM can use such statistics when they are written as sentences is open. A statistic can help in three ways: it can *state* the answer, change the *procedure* by which the model answers, or add *side information* for reading the graph. We separate the three with a graph-blind solver that answers from the statement alone, a structure-free filler that measures what added text does, paired comparisons on the same graphs, truncation as its own outcome, and checks of what each correct answer rests on.

2 Related Work

Structural encodings such as degree and random-walk return probabilities help trained graph models (Dwivedi et al., 2022; Rampášek et al., 2022). LLM accuracy on graph queries depends on the text encoding, the task and the graph, and distracting statements lower it (Fatemi et al., 2024; Wang et al., 2023). Structural descriptions in prompts need not be read as topology (Huang et al., 2024), the modality and order of a graph’s description change answers (Das et al., 2024; Ge et al., 2025), partial-input baselines expose shortcuts (Feng et al., 2019), and added or irrelevant text changes LLM reasoning (Levy et al., 2024; Shi et al., 2023; Liu et al., 2024).

3 Method

Design. Following the work above, we use GraphQA’s tasks, its incident encoding (the most accurate for degree and neighbor queries) and its accuracy scoring; the statistics are the graph-learning encodings—node degree, local clustering (a normalized triangle count) and random-walk return probabilities (RWSE)—stated as sentences; the graph-blind solver is a partial-input baseline; a structure-free *filler*, about as long as the clustering primer, stands for distracting text; and because order and added text change answers, we fix one layout, read every effect against both no primer and filler, and check what responses claim about the graph. Qwen3 runs the same weights with and without thinking (Qwen Team, 2025), so each model is compared with its own reasoning mode. A pilot on the published GraphQA graphs (5–19 nodes) with Gemma 4 and Qwen3-8B/14B found most arms near ceiling without a primer (Appendix C), so the main study uses larger graphs and the smaller Qwen3-1.7B and Qwen3-4B, each in plain and thinking mode (P/T below), zero-shot.

Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	60.50	+7.5	+3.0	+4.0	+10.2	+1.8	-0.5
	1.7T	76.25	+8.8	+1.5	+2.5	+12.2	-1.0	-3.0
	4P	99.25	-6.5	-0.5	-2.2	-8.0	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5	+1.2	+1.0	+2.0	+0.8	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0	+1.0	+0.0	-2.8	+9.2	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.2 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2	-1.2	-1.5	+3.8	-1.2	+1.2
	4T	20.00	+8.5[†]	+24.8[†]	+10.2[†]	+1.8[†]	+24.8[†]	+12.2[†]
Edge exists	1.7P	70.50	+4.0	+3.8	+5.0	+11.5	-8.2	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0	-10.0	-10.0	-2.8	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0
Node count	1.7P	31.50	+6.8	+67.2	+2.0	+47.0	+21.2	+39.8
	1.7T	91.00	+5.5	+9.0	+9.0	+9.0	+1.5	+9.0
	4P	90.25	+8.0	+8.8	+9.2	+9.8	+9.0	+9.2
	4T	99.50	+0.2	+0.2	+0.2	+0.5	+0.0	+0.5
Cycle	1.7P	90.75	-17.5[†]	+8.5	-0.2	+7.0	+2.0	+3.0
	1.7T	97.00	-21.8[†]	-0.5	+2.2	-0.8	-7.5	+3.0
	4P	99.00	-2.2	+0.5	-0.2	-0.2	-2.0	-0.5
	4T	99.25	-5.5	+0.0	+0.2	-0.5	-3.2	+0.5

Table 1: Main-sweep correct shares: no-primer baseline (% of all prompts) and paired change (points) on the same 40-node graphs, $p \leq .50$. Bold: $q < .05$ within (model, task, control); [†]: at least 15% truncation on either side; shaded: the primer states the answer. Neighbor sets: exact set match (set-F1 in Table 4(a)); edge existence: raw correctness (balanced accuracy in Table 4(b)). Node count and cycle check have a constant answer at 40 nodes; truncation shares are in Table 5. P/T: plain/thinking.

Graphs, tasks and primers. Graphs are undirected Erdős–Rényi graphs (Gilbert, 1959) with 40 nodes at edge probabilities $p \in \{.10, .20, .35, .50\}$, and for node degree and edge existence also $\{.65, .75, .85\}$, with 100 graphs per task and density shared by every condition and model; decoding is greedy (temperature 0), one response per prompt, capped at 8,192 tokens (2,048 for the plain models’ dense runs). The incident encoding lists each node’s neighbors on its own line (“Node 3 is connected to nodes 1, 7, 12.”). We use GraphQA’s six tasks: node degree, connected nodes (the queried node’s neighbors), edge count, edge existence, node count and cycle check. At 40 nodes the node count is always 40 and every graph has a cycle, so those two tasks show how a prompt changes the answer rather than how the graph is read. Before the encoding we place no primer, a *degree*, *clustering* or *RWSE* primer (return probabilities after 2 and 3 steps), or *all* three, one sentence per node with values to two decimals; two diagnostics are a sentence stating the number of connected components and the filler, which names every node without any structure (“Node 0 is simply present within the graph G.”; details in Appendix A).

Outcomes and analysis. Each response is **correct**, **wrong** or **truncated** (it used the whole bud-

get), every share has all prompts in its denominator, and an effect is the paired change in the correct share on the same graphs. Every task is scored by exact match, as in GraphQA (for connected nodes, the whole neighbor set); edge existence is also scored by balanced accuracy (Brodersen et al., 2010), because the share of true edges rises from 10% to 85% with density. Paired comparisons use exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs and Benjamini–Hochberg correction within each (model, task, control) family (Benjamini and Hochberg, 1995); q is the adjusted p , and we call an effect significant at $q < .05$. A graph-blind solver answers from the primer text alone (Appendix A). The main sweep comprises 84,000 responses on 700 graphs.

Follow-up runs. Node degree is re-tested with 400 graphs per density for plain and thinking Qwen3-1.7B (no primer, components, clustering, degree, filler), on a fresh seed (1,600 graphs), and on a grid of 20 to 160 nodes at mean degree 8 and 16 (400 graphs per cell, also Qwen3-8B).

4 Results

The graph-blind solver answers node degree and edge count perfectly from the degree primer or all three statistics, by reading the queried degree

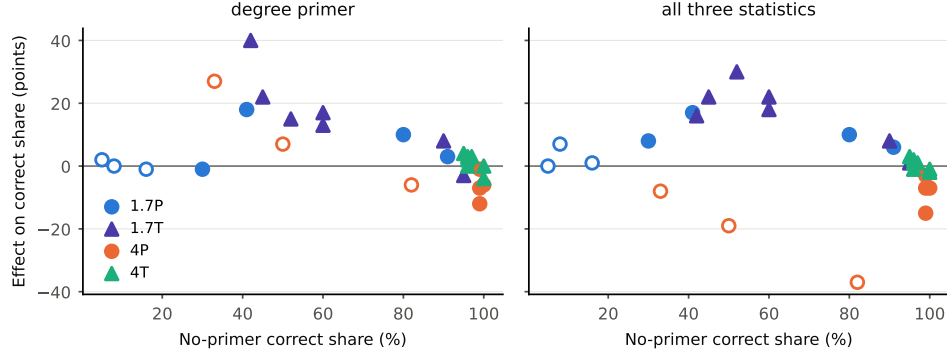


Figure 1: Primers that state the answer, on node degree: effect (points) against the no-primer correct share, per model and density. Left: degree; right: all three statistics (the same degrees in a longer primer). Hollow: at least 15% truncation, or the plain models’ dense runs.

or halving the sum of the stated degrees; among the other primers and tasks whose answer varies it never exceeds 74.6%, against 73.5% with no primer, and on connected nodes 0.8%. These answer-stating cells are shaded in Table 1; we call every other primer–task pair side information.

4.1 Stating the answer

The degree primer raises node-degree accuracy for plain and thinking Qwen3-1.7B by 7.5 and 8.8 points ($q = .0042, .0008$) but costs plain Qwen3-4B, which already counts correctly, 6.5 ($q < .001$). Across models and densities the gain is largest where accuracy without the primer is intermediate and turns negative near ceiling (Figure 1); split-half estimates give the same pattern (+13.9, +16.2), so it is not regression to the mean. A model that has stopped reading the graph is not rescued: in the follow-ups on dense graphs, plain Qwen3-1.7B answers 39 (all other nodes) on 70.8% of finished answers at $p = .85$ even with the degree stated and gains 1.2 points at $p \geq .65$, while its thinking mode gains 24.8 (truncated +9.0).

The stated value replaces or checks the model’s count. Plain Qwen3-4B enumerates the queried node’s neighbors in every response at $p = .50$ without a primer; with it, 62% of responses read the value off, the median response falls from 273 to 34 tokens, and 13 answers break against 1 fixed. When a thinking model reports a conflict between the stated value and its own count, it ends on the stated value in 93% (4B) and 66% (1.7B) of cases. The same values in a longer primer can cost accuracy: inside all three statistics, the degrees cost dense plain Qwen3-4B 21.3 points ($q < .001$) where the degree primer alone gains 9.3 ($q = .03$), and over all densities its look-ups there are right for 88.5%

of queried nodes 0–9 but 53.4% of nodes 10–39. On edge count the degree primer supplies the terms of the sum: plain Qwen3-4B rises from 2.00% to 29.25% (+27.2), the largest gain on a task whose answer varies. Where it states only the size of the answer, on connected nodes, it raises thinking Qwen3-1.7B by 6.2 ($q = .0079$), 3.8 above filler ($q = .18$).

4.2 Side information

Effects of statistics that do not state the answer depend on the statistic and the model: on plain Qwen3-4B’s edge existence clustering gains 4.0 points and RWSE loses 10.0 (balanced accuracy +3.0 and −6.5, $q = .018$ and $.0015$); on its dense node-degree queries clustering gains 11.3 ($q < .001$). Where filler lowers accuracy—15.8 points on plain Qwen3-1.7B’s edge existence (balanced accuracy −10.8), 12.5 on plain Qwen3-4B’s neighbor sets—clustering and RWSE are 19.5 and 20.8 points above filler on the first and 13.5 and 12.5 on the second (all $q < .001$); on plain Qwen3-4B’s edge existence RWSE does no better than filler. The clearest benefit is in what plain Qwen3-1.7B claims: with a statistic it rarely states a queried non-edge as an edge (13.7% of non-edges without a primer, 2.7–4.4% with one), and all three statistics lower its false alarms from 0.51 to 0.33 over all densities and raise its balanced accuracy by 7.8 points ($q = .001$), by 14.9 on dense graphs where without a primer it answers yes to nearly every pair ($p = .0005$). The one re-tested effect, clustering on node degree for plain Qwen3-1.7B at $p \leq .50$, replicates on new graphs (+4.0), a fresh seed (+4.2), the fixed-mean-degree grid (+6.2) and against filler (+3.2, $q = .014$); on dense graphs it is absent, or a loss for thinking Qwen3-1.7B (−5.4, $q = .0015$),

Arm	Primer	Correct	Valid	False		No cycle	
		“yes” (<i>n</i>)	Real cycle	Invented cycle	Not a cycle	All	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	degree	293	13.0	20.5	1.4	65.2	55.6
	all	391	38.4	50.4	7.9	3.3	1.3
1.7T	none	388	79.9	17.8	1.0	1.3	0.5
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	all	385	73.0	21.8	3.1	2.1	1.8
4P	none	396	34.1	23.0	6.8	36.1	25.0
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
	degree	387	13.4	3.9	2.1	80.6	65.6
	all	395	45.1	33.4	6.6	14.9	0.8
4T	none	397	96.5	3.3	0.0	0.3	0.0
	clustering	397	95.5	4.5	0.0	0.0	0.0
	RWSE	398	55.0	2.5	42.5	0.0	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	all	395	92.7	6.3	0.0	1.0	1.0

Table 2: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct* (*n*): count of correct, finished answers; the other columns are % of those *n*, and Real, Invented, Not a cycle and All sum to 100. *Valid*: a cycle whose every step is an edge. *False*: *Invented* has a step that is not an edge; *Not a cycle* is a length-2 or retraced walk. *No cycle*: All names none; *Edge count* argues from the number of edges (valid). Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

and it is absent for Qwen3-8B (Appendix Figure 2).

4.3 Thinking and the budget

Thinking changes more than any statistic that does not state the answer: in the follow-ups on node degree it adds 24.8 points to Qwen3-1.7B without a primer, and thinking Qwen3-4B, near ceiling, moves by at most 2.5 under any primer on node degree, edge existence and neighbor sets. On edge count, thinking Qwen3-4B is right on 97.6% of the answers it finishes but runs out of budget on 79.5% of prompts; the component sentence’s gain there (+24.8) is a fall in truncation (−25.0; Appendix Figure 3).

4.4 Correct is not executed

Every graph has a cycle, so every correct cycle-check answer is “yes”, and Table 2 checks, edge by edge, what each correct answer rests on. Only 36.9% of plain Qwen3-1.7B’s and 34.1% of plain Qwen3-4B’s rest on a real cycle, and 43.5% of plain Qwen3-1.7B’s rest on an invented one. Think-

ing raises the share resting on a real cycle to 80.9% and 96.4% on graphs both modes answer correctly ($q < .05$). With the degree primer the plain models argue instead from the number of edges, a valid argument (55.6% and 65.6%), though on cycle check it costs the 1.7B models 17.5 and 21.8 points, mostly through truncation; RWSE alone leads thinking Qwen3-4B to rest 42.5% of its correct answers on a walk that is not a cycle (length 2 or retraced).

5 Discussion

Substitution, not improvement. Among unflagged cells on tasks whose answer varies, the largest gains come from statistics that state the answer, in the models that compute it least reliably, and a model that already counts correctly loses by switching to a look-up: a primer patches weak execution rather than improving it, and its value depends on the model.

Findable content. More statistics are not better: the same values in a longer primer can cost accuracy, and rounded RWSE takes only 3.30 distinct value pairs per graph at $p = .50$ against 12.50 distinct degrees. Filler shows that layout alone changes answers (Appendix B), so a primer must be judged against structure-free text too.

Measuring execution. Plain Qwen3-1.7B’s raw edge-existence accuracy rises from 54% to 86% between $p = .50$ and $.85$ as its balanced accuracy falls to 0.53, near always answering yes, and set-F1 moves with exact match in only 3 of 5 large contrasts (finished pairs).

6 Future Work

Controlled follow-ups. First, test whether a primer still helps when it cannot directly disclose the answer: omit queried-node values while preserving equivalent statistics for other nodes, and ask multi-step queries that require the incident edges. Second, randomize node labels and preamble order, and compare truly token-matched neutral text under equal generation budgets; these interventions would separate positional and length effects from graph information. Third, use variable-size and deliberately cyclic/acyclic corpora with balanced edge queries, then repeat the paired design on graph families beyond Erdős–Rényi. Finally, measure whether answers jointly respect $d_v = |N(v)|$ and $2|E| = \sum_v d_v$ across queries,

not merely whether each answer matches in isolation. Record intermediate claims only as observable output, and test their consistency against the underlying graph; stated reasoning alone does not establish the procedure that produced a response.

Beyond these tests, our results suggest that a primer helps by doing computation the model cannot do reliably itself: stating the queried node’s degree raises thinking 1.7B by +8.8 points on node degree and lowers plain 4B, which already counts correctly, by −6.5. This points to six broader directions.

Who should compute. Compare four ways of giving a model the same structure: statistics computed in advance and placed in the prompt, graph tools the model can call, code it writes and runs, and the incident encoding alone. Varying model scale, graph size and task would show when structure should be computed for the model and when the model can work it out, a choice every system that pairs an LLM with graph data has to make.

Predicting primer benefit. Our effects track two measurable quantities: how much the primer reveals about the answer, which the graph-blind solver measures, and how poorly the model does without it. A model of the effect as the primer’s information gain times the model’s shortfall, minus a reading cost, could be fitted across many model families and tested on held-out models. The reading cost is substantial: in the dedicated degree follow-ups, filler costs thinking 1.7B 11.4 points at $p \geq .65$. Such a model also needs repeated sampling per query, because most primer effects sit on top of churn: in 89 of 120 (arm, task, primer) cells, a primer changes as many outcomes as filler does.

Graphs that do not fit. Beyond 40 nodes the encoding itself becomes the bottleneck. Under a fixed token budget, which summary keeps the most queries answerable: statistics, sampled subgraphs, or the queried node’s neighborhood?

Searching the primer space. We chose four features and one wording by hand, and both matter. The plain-1.7B clustering gain is 5.01 points on queried nodes in the first third of the node list and −0.11 on the last third, and under RWSE, whose sentences state a return probability “after 2 steps”, 42.5% of thinking 4B’s correct cycle-check answers rest on a walk that is not a cycle (a two-node “cycle” or a retraced path), against 0.0% without a

primer. An automated search over feature, wording, placement and scope, scored against length-matched and shuffled-value controls on held-out graphs, would bound what any primer can achieve.

Trust in stated statistics. Every primer we tested was correct. Among finished degree-primed node-degree responses that report a conflict between the stated degree and the model’s own count, thinking 4B ends on the stated value in 93%, thinking 1.7B in 66% and plain 4B in 56%. Primers corrupted at controlled rates would show whether models verify stated structure or copy it, which matters wherever LLMs rely on metadata that may be stale or wrong.

Mechanism. With open weights, activation patching could show whether a model uses a primer’s content or responds only to a longer, differently formatted prompt.

A controlled benchmark. The controls this study needed could be released as a benchmark, so that future claims that structure helps LLMs face the confounds we met: a graph-blind solver that exposes answer disclosure, truncation as its own outcome, length-matched content-free text, churn-aware paired comparisons, degree-preserving rewiring that changes clustering while fixing the degree sequence, gold answers and prompt length, and checks that correct answers rest on true claims. Without a primer, 43.5% of plain 1.7B’s correct cycle-check answers rest on an invented cycle.

7 Conclusion

Explicit structural statistics raise accuracy on graph queries mainly by supplying the answer, which substitutes for execution rather than improving it: it helps models that cannot compute the answer and costs one that can. Other statistics have smaller, model-specific effects, mostly avoid the cost of structure-free text, and cut the non-edges plain Qwen3-1.7B states as edges. Because correct answers can rest on invented cycles, evaluation must check what an answer rests on, not only whether it is right.

Limitations

We study one model family (Qwen3), 40-node Erdős–Rényi graphs, one encoding and one wording per primer, zero-shot. Node ids 0–9 are both first and single-digit, so node-level results are stated by id.

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A Method Details and Audit Trail

Statistics. With adjacency matrix A and transition matrix $P = D^{-1}A$, the per-node statistics are $d_v = \sum_u A_{vu}$, $c_v = 2t_v/(d_v(d_v - 1))$ and $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$, where t_v counts edges among v ’s neighbors and $c_v = 0$ when $d_v < 2$. Clustering and return probabilities are printed to two decimals, one sentence per node, in node order.

Graph-blind solver. It uses 16 exact rules, 1 heuristic and 8 rules fitted on separate graphs.

Why these features. Degree and RWSE are the structural encodings graph Transformers use (Ram-pásek et al., 2022); local clustering adds how a node’s neighbors connect; the component count is a single global fact that, with the node and edge counts, decides whether a cycle exists. Laplacian encodings are left out because their sign and basis are arbitrary and cannot be stated as facts (Dwivedi et al., 2022).

Detectable effects. In one cell of 100 graphs the median smallest detectable paired effect is 7.4 points; among contrasts without a significant effect (400 paired graphs each), the median detectable gain is 5.7 points for the 1.7B models and 3.2 for the 4B models.

Response-level measures. Rule-based readers extract what each response states: whether it enumerates the queried node’s neighbors or retrieves a stated value, whether it reports a conflict with the primer, and which cycles and edges it asserts, each checked edge by edge against the prompt’s graph.

Coverage. Every result uses an undirected graph with exactly 40 nodes, except the fixed-mean-degree grid (20–160 nodes); each model sees the same graph instances under seven conditions, so pairs are formed on (task, instance, model). Table 3 gives the primer lengths.

Scoring and provenance. A generation that reaches its output cap is *truncated*. Set-F1 on connected nodes is a supplement, scoring a truncated response as zero; changes in balanced accuracy use a paired permutation test with Benjamini–Hochberg adjustment within each model; cells with at least 15% truncation on either side are flagged and left out of pooled summaries. Tagged results come from GraphTalk’s `results-sot` branch (GraphTalk Contributors,

Condition	Mean chars	Information
Components	37	Number of connected components
Degree	891	Degree for each node
Clustering	1,629	Local clustering for each node
RWSE	2,949	Two rounded return values per node
All three	4,691	Degree, clustering and RWSE
Filler	1,829	Node names, no graph statistic

Table 3: Mean preamble length in characters over the main-sweep graphs.

2026): `docs/results/n40-sweep.md`, `docs/results/density-followups.md`, `docs/primer-robustness.md`, `docs/investigate_connections_and_cycles.md`, `docs/primer-directions.md` and the script outputs they cite.

(a) Neighbor-set Set-F1					(b) Edge-existence decisions					
Arm	Primer	None	With	Δ ; q	Arm	Primer	Raw	Bal.	Yes	Trunc.
1.7P	filler	0.9721	0.9624	-0.0096; 0.3114	1.7P	none	0.705	0.799	0.562	0.000
1.7P	components	0.9721	0.9716	-0.0004; 0.9503	1.7P	degree	0.745	0.823	0.517	0.000
1.7P	clustering	0.9721	0.9724	+0.0003; 0.9503	1.7P	clustering	0.743	0.824	0.525	0.000
1.7P	rwse	0.9721	0.9702	-0.0018; 0.9503	1.7P	rwse	0.755	0.827	0.501	0.003
1.7P	degree	0.9721	0.9794	+0.0074; 0.3352	1.7P	all	0.820	0.877	0.448	0.000
1.7P	all	0.9721	0.9626	-0.0094; 0.3846	1.7T	none	0.990	0.990	0.271	0.003
1.7T	filler	0.9732	0.9769	+0.0037; 0.8851	1.7T	degree	0.965	0.976	0.299	0.005
1.7T	components	0.9732	0.9726	-0.0006; 0.9942	1.7T	clustering	0.980	0.986	0.286	0.003
1.7T	clustering	0.9732	0.9830	+0.0098; 0.4254	1.7T	rwse	0.978	0.982	0.283	0.003
1.7T	rwse	0.9732	0.9577	-0.0156; 0.4254	1.7T	all	0.973	0.981	0.295	0.000
1.7T	degree	0.9732	0.9627	-0.0105; 0.4254	4P	none	0.905	0.932	0.357	0.000
1.7T	all	0.9732	0.9733	+0.0001; 0.9942	4P	degree	0.858	0.903	0.407	0.005
4P	filler	0.9871	0.9660	-0.0212; 0.0001	4P	clustering	0.945	0.963	0.323	0.000
4P	components	0.9871	0.9976	+0.0105; 0.0001	4P	rwse	0.805	0.867	0.463	0.000
4P	clustering	0.9871	0.9845	-0.0026; 0.5411	4P	all	0.805	0.867	0.461	0.003
4P	rwse	0.9871	0.9776	-0.0095; 0.0681	4T	none	0.998	0.998	0.268	0.003
4P	degree	0.9871	0.9768	-0.0103; 0.0379	4T	degree	0.998	0.998	0.268	0.003
4P	all	0.9871	0.9743	-0.0128; 0.0209	4T	clustering	0.990	0.993	0.270	0.010
4T	filler	0.9636	0.9720	+0.0084; 0.8279	4T	rwse	0.985	0.987	0.269	0.015
4T	components	0.9636	0.9641	+0.0006; 0.8956	4T	all	0.988	0.992	0.271	0.013
4T	clustering	0.9636	0.9544	-0.0092; 0.8279						
4T	rwse	0.9636	0.9676	+0.0041; 0.8279						
4T	degree	0.9636	0.9685	+0.0050; 0.8279						
4T	all	0.9636	0.9551	-0.0084; 0.8279						

Table 4: Secondary metrics on the same 400 main-sweep prompts per model and condition. (a) Neighbor-set set-F1, with truncations scored zero: each row gives the no-primer baseline, the primer score and paired difference; q adjusts six contrasts per model (paired random-sign tests). (b) Edge-existence raw and balanced accuracy (Bal.) and truncation (Trunc.) use all prompts; yes-rate (Yes) uses finished answers. Compare each primer row with its model’s none row; changes in balanced accuracy are tested in the text (§4). P/T: plain/thinking.

Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 5: Complete main-sweep truncated shares: no-primer baseline (percent of all prompts) and paired changes (percentage points), for the same queries as Table 1. Each of its cells’ *wrong* share equals 100% minus its correct and truncated shares. P/T: plain/thinking; components and filler are diagnostics. No significance marks are assigned to truncation contrasts.

(a) Correct share								
Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	9.67	+0.3	-3.3	-5.3	+2.7	+0.3	-5.3
	1.7T	46.33	+25.7	-9.0	-4.3	+22.7	-1.7	-12.0
	4P	55.00	+9.3	+11.3	+7.7	-21.3	-1.0	+2.0
	4T	97.33	-0.7	-0.7	-0.7	+0.3	-2.0	+0.0
Edge exists	1.7P	77.00	+2.0	+1.3	+1.0	+6.7	-1.0	-1.3
	1.7T	99.00	-0.7	-0.3	+0.3	-0.7	+0.0	-0.7
	4P	90.00	-3.3	+5.3	-0.7	-2.3	-3.3	-6.3
	4T	99.33	+0.7	+0.7	+0.3	+0.3	+0.7	+0.7

(b) Truncated share								
Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	0.33	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
	1.7T	1.67	+9.0	-1.3	-1.3	+0.0	+0.3	-1.0
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.33	+0.3	+0.0	+0.7	+0.7	+0.3	-0.3
Edge exists	1.7P	0.00	+1.3	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4P	0.00	+0.3	+0.0	+0.0	+0.3	+0.3	+0.0
	4T	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0

(c) Edge-existence decisions				
Arm	Primer	Raw	Bal.	Yes
1.7P	none	0.770	0.527	0.987
1.7P	degree	0.790	0.606	0.926
1.7P	clustering	0.783	0.564	0.960
1.7P	rwse	0.780	0.548	0.977
1.7P	all	0.837	0.674	0.907
1.7P	components	0.760	0.507	0.997
1.7P	filler	0.757	0.500	1.000
1.7T	none	0.990	0.993	0.747
1.7T	degree	0.983	0.980	0.753
1.7T	clustering	0.987	0.987	0.750
1.7T	rwse	0.993	0.986	0.763
1.7T	all	0.983	0.975	0.760
1.7T	components	0.990	0.993	0.747
1.7T	filler	0.983	0.984	0.747

Arm	Primer	Raw	Bal.	Yes
4P	none	0.900	0.832	0.803
4P	degree	0.867	0.740	0.870
4P	clustering	0.953	0.932	0.763
4P	rwse	0.893	0.813	0.817
4P	all	0.877	0.756	0.866
4P	components	0.867	0.754	0.849
4P	filler	0.837	0.697	0.873
4T	none	0.993	0.996	0.750
4T	degree	1.000	1.000	0.757
4T	clustering	1.000	1.000	0.757
4T	rwse	0.997	0.998	0.753
4T	all	0.997	0.993	0.760
4T	components	1.000	1.000	0.757
4T	filler	1.000	1.000	0.757

Table 6: Complete dense-extension comparison at $p \in \{.65, .75, .85\}$ for degree and edge existence. (a,b) No-primer correct/truncated share (percent of all 300 prompts), followed by paired primer changes (points); wrong share is the remainder. (c) Edge raw accuracy (Raw) and balanced accuracy (Bal.) include truncations as noncorrect; yes-rate (Yes) uses finished answers. Per-level values are not tested here; the pooled dense contrasts cited in the text are. The plain models’ budget here is 2,048 tokens. P/T: plain/thinking.

B Additional Results

Presentation effects. Filler, a preamble that adds no information and only names the nodes the encoding already lists, also changes answers: a question filler flips is flipped by a primer 38.5–69.1% of the time, against 3.1–15.2% of other questions. Under filler, plain Qwen3-1.7B largely stops listing node ids on node count (59.5% → 12.2% of answers), states more non-existent edges (+9.6), plain Qwen3-4B invents more neighbors (+14.8), and thinking Qwen3-4B runs out of budget less often on edge count (78.2% → 65.0% truncated while working), matching the sensitivity of LLM reasoning to added and irrelevant text (Levy et al., 2024; Shi et al., 2023). Regenerating identical prompts for plain Qwen3-1.7B on node degree changes 5.3% of outcomes.

Clustering on node degree. The gain for plain Qwen3-1.7B (Figure 2) falls on queried nodes with low ids (+5.01, +0.75 and −0.11 points by thirds over all seven densities; OLS coefficient −1.74 per 10 ids, $p = .018$), where filler shows no such pattern, and its net gain is in copying errors (224

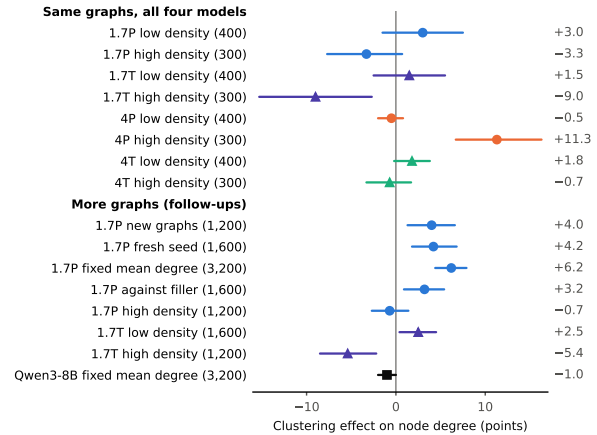


Figure 2: The clustering primer on node degree: effect in points with 95% bootstrap intervals; graphs per row in brackets. Top: all four models on the main-sweep graphs at low ($p \leq .50$) and high density ($p \geq .65$). Bottom: follow-ups with more graphs for plain Qwen3-1.7B, where a screen found the effect, its thinking mode, and Qwen3-8B (black squares) on the fixed-mean-degree grid ($n = 20$ –160).

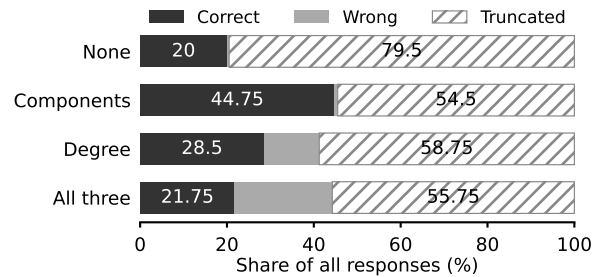


Figure 3: Thinking Qwen3-4B on edge count: correct, wrong and truncated shares of *all* responses. Conditions from top: none, component count, degree, all three.

corrected, 176 introduced, all densities). In the thinking mode it is +2.5 at low density and −5.4 at high ($q = .043, .0015$).

Thinking. Thinking costs 17.1 times the tokens of the plain mode at $p = .10$ and 11.5 times at $p = .85$. Plain models run out of budget in loops (83.9% and 89.6% of their truncated responses), thinking models while still working. Under the degree primer, thinking Qwen3-1.7B’s first candidate answer appears earlier in its trace (median 27% of the way through, against 44% without a primer).

Reading errors. Plain Qwen3-4B’s wrong look-ups under the degree primer copy the adjacent line’s degree more often than chance (17 vs 10.4). Plain Qwen3-1.7B’s edge-existence false alarms rise with the neighbors the two nodes share (1.7% with none, 65.9% with four or more). Plain Qwen3-

1.7B’s node-degree lists miss a neighbor in 24.6% of cases and add one in 3.3%, and 93.0% of plain Qwen3-4B’s missed neighbors lie below the queried node’s id, against 43% by chance. For plain Qwen3-4B, clustering is the only primer that lowers edge-existence false alarms ($0.16 \rightarrow 0.08$).

C Screening and Focused Robustness Checks

Earlier screens. A pilot on the published GraphQA graphs (5–19 nodes) evaluated Gemma 4 E4B/12B and Qwen3-8B/14B, each in plain and thinking mode; most arms were near ceiling without a primer on most tasks.

Selection and replication. The clustering-on-degree condition emerged from a 40-cell screen: 700 dedicated graphs overlap it, while 2,100 do not. The independent fresh seed gives +4.2 points ($p = .00055$), with screen-wide Bonferroni-adjusted $p = .022$.

Sensitivity. Split-half means remain +13.9 and +16.2 points for answer-carrying cells at intermediate baselines, and the band means hold when the truncation flag is set at 10% or 50%.

Fixed-size effect heterogeneity. Table 7 compares primer effects between density groups; `robustness_audit.py` regenerates it from saved responses.

Direct bundle contrast. Across 300 paired dense plain-4B degree queries, clustering alone is correct on 66.33% versus 33.67% with *all* (-32.67 points, $q < .001$ across six alternatives).

Arm / primer	Low	High	High–low	q
1.7P / cluster.	+3.81	−0.67	−4.48	.00445
1.7T / cluster.	+2.50	−5.42	−7.92	< .001
1.7T / degree	+11.31	+24.75	+13.44	< .001

Table 7: Primer effects on degree correct share (points) relative to no primer. Low: $p \leq .50$ (1,600 pairs); high: $p \geq .65$ (1,200 pairs). P/T: plain/thinking.